📊 Forecasting Apple Stock with a Deep Learning Model

🔍 Overview

This project builds and evaluates a neural network-based forecasting model to predict Apple Inc. (AAPL) closing prices. It uses historical market data combined with technical indicators to train a feedforward neural network. The system is capable of:

- Predicting the next day's price based on the most recent 5-day window
- Evaluating its performance using multiple metrics (MAE, RMSE, MAPE, Directional Accuracy)
- Forecasting five days into the future
- Visualizing predictions alongside real stock trends

💾 Data Source & Feature Engineering

🧠 Data Retrieval

- Source: Yahoo Finance
- Ticker: `AAPL`
- Time Range: Last 5 years
- Raw Features: `Open`, `High`, `Low`, `Close`, `Volume`

🛠️ Technical Indicators

Using the `ta` library, the model includes key indicators:

- Momentum: RSI, Stochastic, MACD
- Volatility: Bollinger Bands, ATR
- Trend: EMA
- Volume-based: OBV

These indicators are engineered as input features to improve the model's understanding of price dynamics.

🧱 Modeling Pipeline

🪜 Step-by-Step Flow

1. Create 5-day input windows using the selected indicators.
2. Scale features and target (close price) separately using MinMaxScaler.
3. Split into training and test sets (80/20).
4. Train a feedforward neural network with:
    - 256-128-1 architecture
    - ReLU activations
    - Adam optimizer
5. Predict the next close price and evaluate performance.

📊 Evaluation Metrics

Metric	Description
MAE	Average absolute dollar error
RMSE	Penalizes larger errors more than MAE
MAPE	Mean percentage error (scale-invariant)
Directional Accuracy	% of time the model correctly predicted price movement (up/down)

🔢 Example Output

```
Test MAE ≈ $3.22
RMSE: $4.12
Directional Accuracy: 97.99%
MAPE: 2.74%

Sample Predictions vs Actual:
  Actual: $174.23 | Predicted: $171.88 | Error: 1.35%
  Actual: $175.14 | Predicted: $176.42 | Error: 0.73%
  ...
```